Replication package

Can Words Move Inflation Expectations? A Counterfactual-Text Laboratory

Authors: Fernando Barros Jr., Fabio Y. S. Motoki, Valdemar Pinho Neto, and Victor Rangel
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Contact: victorrsr@al.insper.edu.br

This is the isolated replication package for the current paper. The complete reported specification uses GPT-5.5 for the original-minute calibration paths, counterfactual rewrites, and synthetic-forecaster experiments. It does not read from the parent research workspace, require a provider credential, or invoke a remote model during exact replication.

The package follows the AEA Data and Code Availability guidance: a single master script runs the reported analysis; every frozen input has a SHA-256 checksum; every active exhibit is mapped to its producing program; public and author-generated data are documented separately; R dependencies are locked; and automatic gates check the paper's main numerical claims.

One-command reproduction

Start in the package root, the directory containing this README.

```
Rscript code/00_setup.R
Rscript code/00_master.R
```

00_setup.R restores the versions in renv.lock, tests the installed rstan stack, and recompiles the locked Stan stack only if the compatibility test fails. 00_master.R verifies the frozen inputs, creates an expendable working copy of the SQLite database, constructs the GPT-5.5 calibration dataset, runs all analysis and reporting stages, and applies the final verification gates. It never alters data/generated/prompt_iv_replication_source.sqlite.

After an interrupted run, resume completed stages with:

```
Rscript code/00_master.R --resume=true
```

A successful run leaves status=pass in all 87 rows of results/verification_report.csv. Step logs and elapsed times are written under logs/.

Scope of reproducibility

The default reproduction begins from frozen generated texts, blinded scores, embeddings, and synthetic-forecaster responses. This is the correct starting point for exact replication because a future remote-model response is not guaranteed to be byte-identical after provider updates.

The package has two reproducibility layers:

1. **Exact analysis reproduction.** code/00_master.R produces all 29 active LaTeX tables and 10 active PDF figures in data/metadata/exhibit_manifest.csv, plus their machine-readable inputs.
2. **Collection transparency.** code/collection/ contains the versioned prompts, task constructors, quality gates, database schema, and runner code. Collection is not called by the master; its example configuration is a dry-run and needs explicit opt-in before any remote request.

The package does not claim byte-exact recollection from a future model endpoint. The published results are exactly reproducible from the distributed frozen data.

Priors, arms, and regenerations

Use meeting index t, forecaster profile $i \in \{A, U\}$, textual arm j, and regeneration index s. For each meeting and profile, the original-minute run constructs a profile-specific pre-treatment prior P_t^i along the chronological path of published, unmodified minutes. Meeting 203 begins at 3.8 percent; thereafter the prior for profile i is that profile's forecast after the preceding original minute.

Before any counterfactual at meeting t is evaluated, P_t^i is frozen. The same value is supplied to every arm j and every regeneration s in the meeting-profile cell. The experimental revision is

$$\Delta E^i_{tjs} = E^i_{tjs} - P^i_t.$$

"Common prior" therefore means common across arms and regenerations within a meeting and profile. It does not mean $P^A_t = P^U_t$. Nor does a shared value of s connect two meetings: regeneration 1 at meeting t is not compared with regeneration 1 at meeting $t-1$. The two profile revisions are combined with estimated predictive calibration coefficients, not averaged. Intercept and meeting controls are fixed within counterfactual comparisons and cancel.

The separately labelled arm-specific recursive exercise answers a different question. In that exercise, each arm-regeneration-profile combination is a trajectory, and its forecast at meeting $t-1$ is supplied as its prior at meeting t. The analysis scripts verify both the common-prior invariant and the recursive transitions before estimation. See the two collection protocols under documentation/.

GPT-5.5 calibration

At run time, code/01_data_preparation/build_gpt55_calibration_data.R constructs outputs/work/gpt55_calibration_data.dta. It reads the complete 142-observation original-minute panel (71 meetings by two profiles) from run common_prior_original_v1_20260712, requires model gpt-5.5, reasoning effort high, completed status, and the expected prompt/version fields, and verifies the forecast-revision identity and profile-specific chronological continuity.

Those original-minute forecast levels are joined to the public aggregate Focus twelve-month-ahead IPCA median, original-minute tone and word count, and the frozen meeting-level activity controls. The calibration coefficients are then estimated once on the 71 meetings and held fixed in the text experiment. Thus calibration and experimental outcomes use the same model vintage and profile definitions while remaining separate statistical stages.

Computational requirements

The validated environment is Windows 11 (64 bit), R 4.5.2, 32 GB RAM, a C++ toolchain compatible with R 4.5 and rstan, a Cairo-capable graphics device, and at least 5 GB free disk space. No GPU is required. Internet access is needed only for the first dependency restore.

The scripts request single-threaded build, Stan, data-table, and text-analysis behavior where supported to bound memory use. Reserve about 90 minutes for a first setup without a compatible package cache; later analysis-only runs are substantially faster. Exact accepted timings are recorded in logs/master_run_times.csv and summarized in documentation/CLEAN_ROOM_VALIDATION.md.

Validation status (2026-08-16): PASS. The accepted clean-room run passed all 76 frozen-input checks, completed all 15 master stages, passed all 87 artifact and numerical gates, reproduced all 29 tables, and verified all 10 figures.

Directory map

Path	Role
code/00_setup.R	Restores the locked R environment.
code/00_master.R	Runs the complete exact replication.
code/01_data_preparation/	Constructs the run-time GPT-5.5 calibration input and package metadata.
code/02_analysis/	Reported econometric and validation analysis.
code/03_reporting/	Publication-table and figure construction.
code/04_verification/	Input, result, and package-manifest checks.
code/collection/	Auditable LLM collection layer; excluded from the master.
data/raw/	Public Copom minutes and frozen non-LLM calibration controls.
data/analysis/	Meeting/cell-level analysis inputs.
data/generated/	Immutable SQLite database of frozen generated data.
data/metadata/	Provenance, dictionaries, checksums, inventories, and exhibit map.
outputs/	Machine-readable intermediate estimates and temporary work files.
results/	Reproduced exhibits, numerical macros, and QA reports.
reference/	Frozen author-side copies of the current exhibits.
documentation/	Public protocols, data statement, and validation records.
logs/	Per-step logs, timings, and session information.

Primary analysis inputs

- data/generated/prompt_iv_replication_source.sqlite (about 200 MB): frozen generated texts, 15,762 completed selected synthetic-forecaster results, scores, embeddings, public external-series extracts, run metadata, and audit inventories.
- data/raw/calibration_controls/copom_meeting_activity_controls.csv (71 meetings by five fields): meeting/date keys and frozen agriculture, industry, and services activity-index levels used only as calibration controls.
- data/analysis/fiscal_sample_flags_by_meeting_20260630.csv (71 meetings by three fields): RTN-window and CDS-series-availability flags; no CDS level is redistributed.
- data/analysis/realized_dose_dataset.csv (2,130 text cells by 29 fields): lexicon, blinded-semantic, and pairwise dose measures.
- data/raw/copom_minutes/ (71 files): public minutes for meetings 203--273.

The calibration .dta is a generated work file, not a distributed input. Detailed field definitions are in the metadata dictionaries. The database table referee_human_audit_ratings has zero rows by design: a blinded packet was prepared, but no human ratings were collected, and no human-audit estimate is imputed.

Data provenance and access

Only public data and author-generated synthetic data are used; there is no confidential institution-level Focus microdata. Source URLs, retrieval fields, coverage, redistribution status, citations, and included-file mappings are in data/metadata/data_sources.csv and data/metadata/external_source_manifest.csv.

Distributed third-party data come from Banco Central do Brasil Copom minutes, Focus/Olinda aggregate and Top-5 statistics, BCB SGS series, and Tesouro Nacional RTN publication dates. Historical World Government Bonds CDS values and the Ipeadata/J.P. Morgan EMBI series are not redistributed because clear redistribution permission was not identified. The active fiscal table uses only a binary CDS-coverage flag; EMBI values are not read by the master.

The three activity-control columns were extracted without transformation from an author-maintained workbook. The retained workbook metadata identifies BCB/Depec and monthly index levels but does not preserve SGS identifiers, seasonal-adjustment flags, or the original download date. The public package therefore freezes and checksums the exact 71-row extract and discloses this provenance limit instead of asserting an unrecoverable series vintage. Details and hashes are in data/raw/calibration_controls/README.md.

The main frozen GPT-5.5 run identifiers are:

Purpose	Run ID	Tasks/results
25-regeneration rewrite corpus	v10_referee_d5_25draws_20260620	17,750 rewrites
Original path and calibration forecasts	common_prior_original_v1_20260712	142 forecasts
Common-prior binary Prompt-IV	common_prior_main_25draws_v1_20260712	7,100 forecasts
Common-prior six-arm dose response	common_prior_dose_6arm_5draws_v1_20260713	4,260 forecasts
Arm-specific recursive paths	arm_specific_prior_dose_6arm_5draws_v1_20260714	4,260 forecasts

Analysis workflow

The 15 master stages are:

1. verify the 76 frozen inputs;
2. construct and validate the GPT-5.5 calibration dataset;
3. estimate the common-prior Prompt-IV and dose/holdout/fiscal checks;
4. run forecast-only out-of-sample validation;
5. estimate arm-specific recursive paths;
6. run identification, placebo, regime, and wild-cluster checks;
7. estimate Bayesian and empirical-Bayes calibration alternatives;
8. run external, Top-5, dispersion, moment, and orthogonality checks;
9. run the sunspot/perceived-law diagnostic;
10. estimate the proper-prior posterior model tournaments;
11. evaluate structural-topic preservation;
12. evaluate embeddings and alternative tone instruments;
13. build integrated tables and figures;

14. build baseline figures; and
15. verify all results and exhibits.

The exact exhibit-to-program map is data/metadata/exhibit_manifest.csv. Intermediate estimates remain under outputs/tables/, so each publication entry can be traced to a machine-readable result.

Randomness and verification

All local simulation, bootstrap, topic-model, and Bayesian steps set explicit seeds. The main seed is 20260715; the common-prior bootstrap seed is 20260714. Deterministic targets use tight tolerances; stochastic posterior summaries use documented MCMC tolerances.

The final gate verifies the headline Prompt-IV estimate and standard error, reduced form and first stage, regeneration-specific signs, ordered dose means, strict held-out estimate, calibration and forecast diagnostics, placebos, wild-cluster inference, regime/sunspot/embedding results, exact normalized text for all 29 tables, and existence/nonzero size for all 10 figures. It also writes results/intext_results.csv and results/intext_numbers.tex as the prose-value source of truth.

Collection-code boundary and security

Exact reproduction needs neither Python nor provider access after the R environment is restored. Collection code is retained only to audit the frozen inputs. Remote calls cost money, can change over time, and are never made by the master. No credential is stored in the package; the example configuration uses runner="dry-run", allow_real_llm=false, and one worker. Any new collection must receive a new run ID and must not overwrite the frozen database.

Licenses and deposit checklist

Code is BSD 3-Clause (LICENSE). Author-generated data are CC BY 4.0 (LICENSE_DATA). Third-party documents/data remain subject to provider rights and are not relicensed by the authors.

Before deposit, the corresponding author should upload this directory rather than its parent workspace, rerun the master in the uploaded copy, confirm all verification rows pass, add the assigned repository identifier and final article citation, and retain the top-level README files, licenses, lockfile, and checksum manifests. For GitHub, the SQLite database must use Git LFS as specified in .gitattributes; the repository mirror is not a substitute for a durable archive DOI.